

TMTF

Testy McTestFace

Let's play a game!

- There is a pattern that describes sequences of numbers
- You will guess sequences of numbers, and I will tell you whether they fit the pattern
- When you think you know the pattern, you can guess it
- I will start you off with a sequence of numbers that fits the pattern

2 4 6 fits the pattern

...

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The trick is that *any sequence of increasing numbers fits the pattern*

If our hypothesis is "even numbers", then knowing 4 6 8 fits the pattern doesn't tell us anything new

If our hypothesis is "numbers that differ by 2", then 5 7 9 does *not* confirm that this is true; just that it *might* be true

What inputs really help us zero in on the pattern?

Based on Peter Wason's 1960 psychology study on confirmation bias

From Wikipedia: Confirmation bias (also confirmatory bias, myside bias, [\[a\]](#) or congeniality bias) is the tendency to search for, interpret, favor, and recall information in a way that confirms or supports one's prior beliefs or values.

How does this relate to testing?

- Avoid additional tests that confirm what you already know
- Test cases that pass *do not* reveal errors in your code
- Test cases that fail reveal a flaw (could be a faulty test case)
- Test boundaries
 - 0, length-1, length, and so on
- Test "corner" cases
 - empty string, null, 0,

TMTF (Testy McTestFace)

Really simple software for testing "mystery" Java functions until you think you know what it does

How to run

- Download the appropriate version of TMTF
- Double click to run it
- Download a json file
- File => Load json
- Enter test cases, use ✓ to run tests
- When you have are confident you know what the method does, describe it in the box

